

An odd-window Gabor frame on a non-lattice set of density two

Candidate counterexample packet

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Status: candidate counterexample, likely valid pending expert review.

Abstract

Faulhuber asks whether the Lyubarskii–Nes non-frame theorem for odd windows and lattices of density $(n+1)/n$ extends to arbitrary relatively separated discrete sets of the same lower Beurling density. We answer the proposed universal extension negatively already for $n = 1$. An explicit odd C_c^∞ window g and the non-lattice periodic set

$$\Gamma = \mathbb{Z}^2 \cup ((1/4, 0) + \mathbb{Z}^2)$$

give a Gabor frame for $L^2(\mathbb{R})$, while $D^-(\Gamma) = D^+(\Gamma) = 2$. The proof is exact: the Zak transform of g vanishes on two vertical lines, and the quarter-translate moves those lines to a disjoint pair. The two-window Zak multiplier is consequently strictly positive on the compact torus.

1 Source question

The source is M. Faulhuber, *On the Parity under Metaplectic Operators and an Extension of a Result of Lyubarskii and Nes*, arXiv:1901.01220v2 [1]. On PDF page 9, after extending the odd-window non-frame theorem from separable lattices to all lattices in dimension one, the paper asks:

Does the result in Corollary 1.2 depend on the group structure of Λ or does the result hold for any relatively separated, discrete set $\Gamma \subset \mathbb{R}^2$ with lower Beurling density $(n+1)/n$?

This insight leads to the following question: Does the result in Corollary 1.2 depend on the group structure of Λ or does the result hold for any relatively separated, discrete set $\Gamma \subset \mathbb{R}^2$ with lower Beurling density $\frac{n+1}{n}$? At the moment, the author has no clue in either direction and, therefore, considers the question as an interesting open problem.

Figure 1: The open questions in arXiv:1901.01220v2, PDF page 9.

The question is naturally universal in n and in the odd window. The construction below refutes that universal extension at $n = 1$. It does not settle analogous assertions for each fixed $n \geq 2$, nor either of the two higher-dimensional questions in the source.

2 Definitions and statement

For $x, \omega \in \mathbb{R}$, set

$$T_x f(t) = f(t - x), \quad M_\omega f(t) = e^{2\pi i \omega t} f(t).$$

For a discrete $\Gamma \subset \mathbb{R}^2$, the associated Gabor system is

$$\mathcal{G}(g, \Gamma) = \{M_\omega T_x g : (x, \omega) \in \Gamma\}.$$

It is a frame if there are $0 < A \leq B < \infty$ such that

$$A\|f\|_2^2 \leq \sum_{(x, \omega) \in \Gamma} |\langle f, M_\omega T_x g \rangle|^2 \leq B\|f\|_2^2 \quad (f \in L^2(\mathbb{R})).$$

Define

$$g(t) = \begin{cases} t \exp\left(-\frac{1}{1-4t^2}\right), & |t| < \frac{1}{2}, \\ 0, & |t| \geq \frac{1}{2}, \end{cases} \quad (1)$$

and

$$\Gamma = \mathbb{Z}^2 \cup ((1/4, 0) + \mathbb{Z}^2). \quad (2)$$

Theorem 1 (Counterexample to the irregular-set extension). *The function g in (1) is odd and belongs to Feichtinger's algebra $S_0(\mathbb{R})$. The set Γ in (2) is relatively separated and discrete, is not a lattice, and satisfies*

$$D^-(\Gamma) = D^+(\Gamma) = 2.$$

Nevertheless, $\mathcal{G}(g, \Gamma)$ is a frame for $L^2(\mathbb{R})$. Hence the first open question above has a negative answer as a universal extension: the Lyubarskii–Nes obstruction does depend on more than lower Beurling density.

3 Proof intuition

At the critical lattice $\mathbb{Z}^2$, the Zak transform turns a multi-window Gabor frame operator into multiplication by the sum of the squared Zak transforms of its windows. Thus two windows give a frame exactly when their Zak transforms never vanish simultaneously.

The compact support of (1) makes its Zak transform completely transparent. On one half of a fundamental interval only the translate $k = 0$ contributes, and on the other half only $k = 1$ contributes. Since g vanishes only at its center and endpoints, its Zak transform has precisely the vertical zero lines $x = 0$ and $x = 1/2$ modulo one. Translating g by $1/4$ moves these to $x = 1/4$ and $x = 3/4$. The two pairs are disjoint, so the sum of squares has a positive minimum. The two windows over $\mathbb{Z}^2$ are exactly the single-window Gabor system over the two-coset set Γ .

4 Zak-transform lemma

Use the convention

$$\mathcal{Z}f(x, \omega) = \sum_{k \in \mathbb{Z}} f(x - k) e^{2\pi i k \omega}. \quad (3)$$

The Zak transform extends to a unitary map from $L^2(\mathbb{R})$ onto the usual quasi-periodic realization of $L^2([0, 1]^2)$.

Lemma 2 (Critical multi-window criterion). *Let $h_1, \dots, h_m \in S_0(\mathbb{R})$. Then*

$$\bigcup_{j=1}^m \mathcal{G}(h_j, \mathbb{Z}^2)$$

is a frame if and only if there are $0 < A \leq B < \infty$ such that

$$A \leq \sum_{j=1}^m |\mathcal{Z}h_j(x, \omega)|^2 \leq B \quad (4)$$

on $[0, 1]^2$. In particular, because these Zak transforms are continuous, the union is a frame if they have no common zero.

Proof. Under $\mathcal{Z}$, the functions $M_\ell T_k h_j$ become, up to a unimodular factor,

$$e^{2\pi i(k\omega - \ell x)} \mathcal{Z} h_j(x, \omega).$$

Parseval's identity for the Fourier basis $\{e^{2\pi i(k\omega - \ell x)}\}_{k, \ell \in \mathbb{Z}}$ shows that the frame operator is unitarily equivalent to multiplication by the function in (4). Such a multiplication operator is bounded and invertible exactly when its multiplier is bounded above and away from zero. The continuity assertion follows from $S_0(\mathbb{R}) \subset W_0(\mathbb{R})$; alternatively it is immediate for the particular compactly supported smooth windows used below. This standard Zak diagonalization is also written out for periodic configurations in [3]. $\square$

5 Proof of Theorem 1

Proof. The bump in (1) is smooth at $\pm 1/2$ with all derivatives zero there. Hence $g \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R}) \subset S_0(\mathbb{R})$. It is odd, and

$$g(t) \neq 0 \quad \text{for } 0 < |t| < 1/2. \quad (5)$$

We first determine the exact zero set of $\mathcal{Z}g$ in a fundamental domain. If $0 < x < 1/2$, only $k = 0$ contributes to (3), whereas if $1/2 < x < 1$, only $k = 1$ contributes. Therefore

$$\mathcal{Z}g(x, \omega) = \begin{cases} g(x), & 0 < x < 1/2, \\ e^{2\pi i\omega} g(x - 1), & 1/2 < x < 1. \end{cases} \quad (6)$$

At $x = 0$ the only possible interior contribution is $g(0) = 0$, and at $x = 1/2$ the two endpoint contributions vanish. Combining this with (5) gives

$$\{(x, \omega) \in [0, 1]^2 : \mathcal{Z}g(x, \omega) = 0\} = (\{0, 1/2\} \times [0, 1]). \quad (7)$$

Translation by a acts on the Zak transform by

$$\mathcal{Z}(T_a g)(x, \omega) = \mathcal{Z}g(x - a, \omega).$$

For $a = 1/4$, the zero set modulo one is consequently

$$\{1/4, 3/4\} \times [0, 1),$$

which is disjoint from (7). Thus the continuous function

$$Q(x, \omega) = |\mathcal{Z}g(x, \omega)|^2 + |\mathcal{Z}(T_{1/4}g)(x, \omega)|^2$$

is strictly positive on the compact torus. It has a positive minimum and a finite maximum. Lemma 2 proves that

$$\mathcal{G}(g, \mathbb{Z}^2) \cup \mathcal{G}(T_{1/4}g, \mathbb{Z}^2)$$

is a frame. Since translations commute,

$$M_\ell T_k(T_{1/4}g) = M_\ell T_{k+1/4}g,$$

so this multi-window system is exactly $\mathcal{G}(g, \Gamma)$.

It remains to check the geometry of Γ . It is the union of two distinct cosets of $\mathbb{Z}^2$, hence is uniformly discrete (the minimum separation is $1/4$) and therefore relatively separated. Every unit-area fundamental cell contains two points up to boundary conventions, so standard periodic counting gives $D^-(\Gamma) = D^+(\Gamma) = 2$. Finally, $0, (1/4, 0) \in \Gamma$ but $(1/2, 0) \notin \Gamma$; thus Γ is not an additive subgroup and hence is not a lattice. All hypotheses in the source question at $n = 1$ are met, while the proposed non-frame conclusion fails. $\square$

6 Verification, scope, and novelty status

The proof is analytic and does not depend on computation. The accompanying script samples the exact piecewise Zak multiplier, checks the four vertical zero classes, and reports a positive numerical lower bound as a regression test. This is a sanity check, not part of the proof.

The counterexample settles the first source question negatively in its universal form, already at $n = 1$. It does not decide whether a corresponding non-frame theorem might hold at each fixed density $(n + 1)/n$ for $n \geq 2$, and it does not address the higher-dimensional questions.

A bounded novelty search covered the run’s ledger/indexes and arXiv searches for the source title, “odd Gabor window”, “relatively separated”, “lower Beurling density”, “Lyubarskii–Nes”, compactly supported odd windows, and unions of shifted integer lattices. It also inspected the closely related later paper arXiv:2502.09510 [3]. That paper uses the same standard Zak multiplier for periodic configurations and proves generic frame statements under a finite separated-zero hypothesis, but the search did not find the explicit window (1), the two-coset counterexample (2), or an explicit answer to the 2019 question. Novelty confidence is therefore moderate, not bibliographically certified; the mechanism is elementary and close to known Zak-transform technology.

Human-review recommendation. Prioritize the identification of the source quantifiers, the critical multi-window Zak criterion, and the exact endpoint zero calculation in (6)–(7). These are the only substantive points on which the counterexample depends.

References

- [1] M. Faulhuber, *On the Parity under Metaplectic Operators and an Extension of a Result of Lyubarskii and Nes*, Results Math. 75 (2020), article 8; arXiv:1901.01220v2.
- [2] Y. Lyubarskii and P. G. Nes, *Gabor frames with rational density*, J. Fourier Anal. Appl. 19 (2013), 805–822; arXiv:1108.2684.
- [3] M. Faulhuber, *Gabor systems with Hermite functions of order n and oversampling greater than $n + 1$ which are not frames*, Sampling Theory, Signal Processing, and Data Analysis (2025); arXiv:2502.09510v2.